

ENGINEERING THERMODYNAMICS (MME 365)

Kofi Owura Amoabeng (PhD)

Department of Mechanical Engineering

Kwame Nkrumah University of Science and Technology, Kumasi, Ghana

Course Content

- Introduction (Definitions and Concepts)
- First Law of Thermodynamics
- Properties of Liquids and Vapours
- Perfect Gases
- Second Law of Thermodynamics
- Vapour and Gas Power Cycles
- Refrigeration Cycles

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References/Reading List /Recommended Textbooks

- Thermodynamics: An Engineering Approach by Yunus Cengel and Michael Boles. McGraw-Hill Education; 7th Edition.
- Fundamentals of Engineering Thermodynamics by M. J. Moran, H. N. Shapiro. 5th Edition.
- Engineering Thermodynamics by R. K. Rajput, 3rd Edition.
- Lecture Slides and Other thermodynamics textbooks available.

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ENGINEERING THERMODYNAMICS (MME 365)

INTRODUCTION

(Basic Concepts and Definitions)

Kofi Owura Amoabeng (PhD)

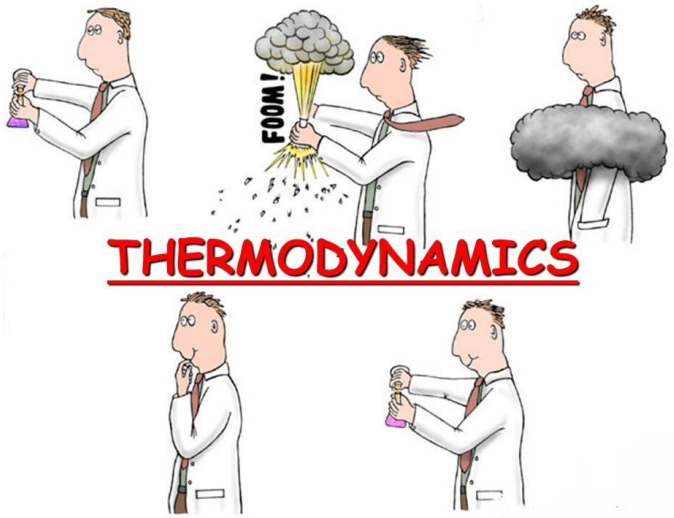
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Introduction

What is thermodynamics?

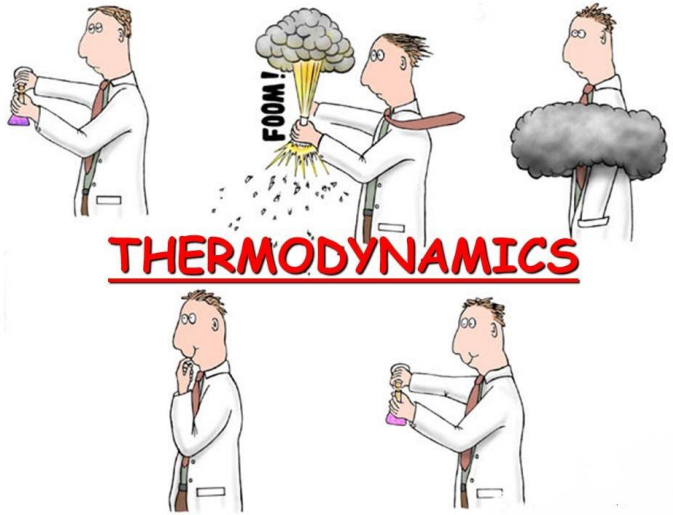
- **Thermodynamics is concerned with the study of energy, its forms and transformations, and the interactions of energy with matter.**
- Thermodynamics is both a branch of physics and an engineering science.
- Engineers are generally interested in studying systems and how they interact with their surroundings. To facilitate this, engineers extend the subject of thermodynamics to the study of systems through which matter flows.



Introduction

Two Approaches to Thermodynamics

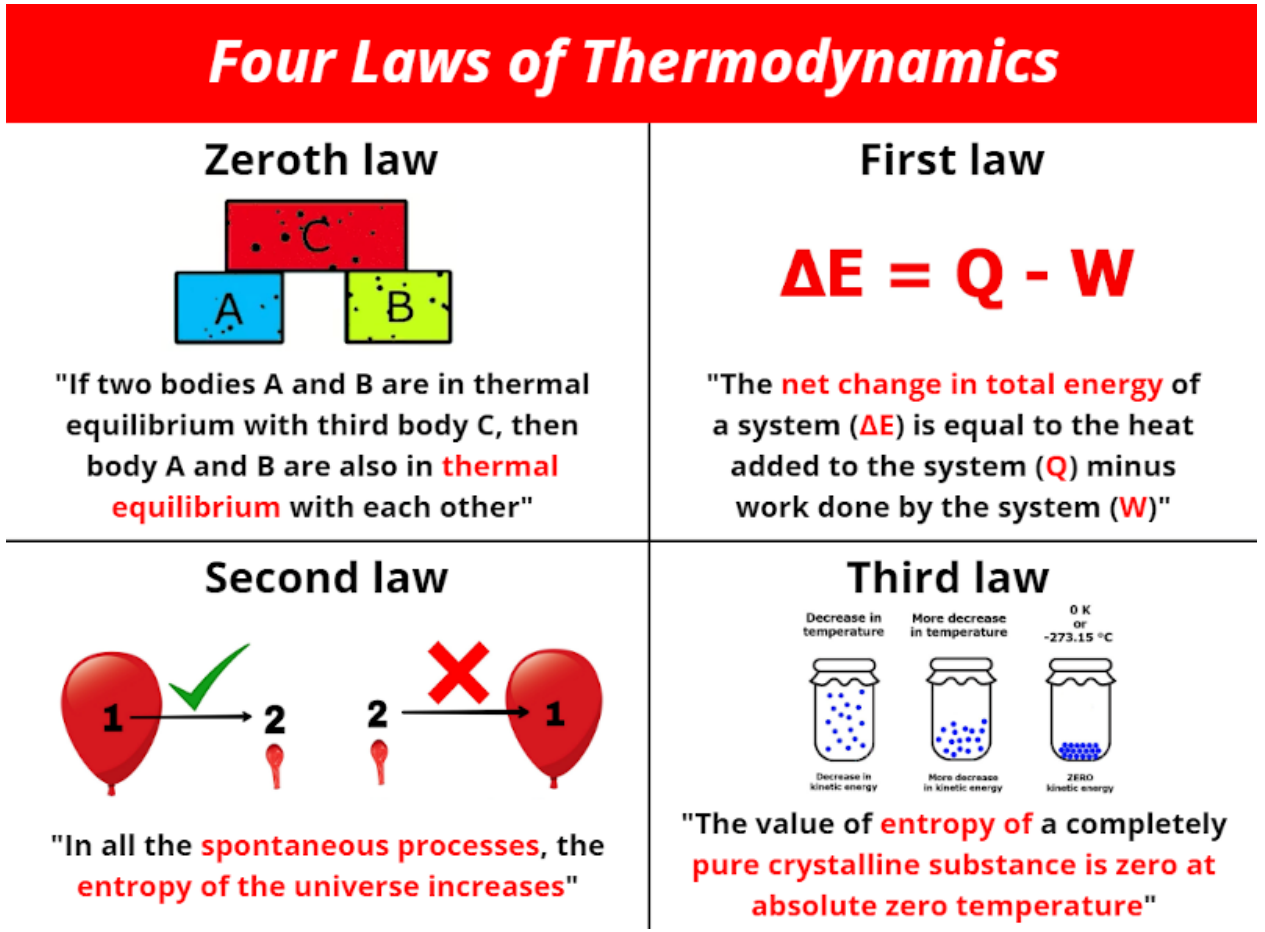
- Systems may be studied from the macroscopic or microscopic approach.
- The **macroscopic approach** to thermodynamics is **concerned with the gross or overall behaviour**. This is sometimes called classical thermodynamics.
- The **microscopic approach**, is concerned directly with the structure of matter. It is known as statistical thermodynamics.
- The macroscopic viewpoint is what is generally adopted in engineering thermodynamics but for understanding concepts are interpreted from the microscopic viewpoint.



Introduction

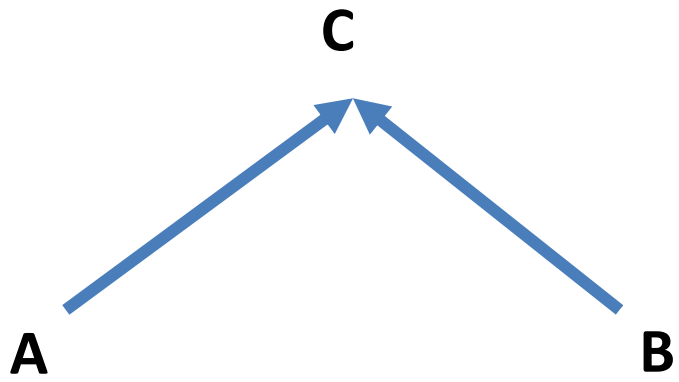
Laws of Thermodynamics

- Four laws of thermodynamics are:
- ✓ Zeroth law
- ✓ First law
- ✓ Second law
- ✓ Third law
- The state postulate



Zeroth Law of Thermodynamics

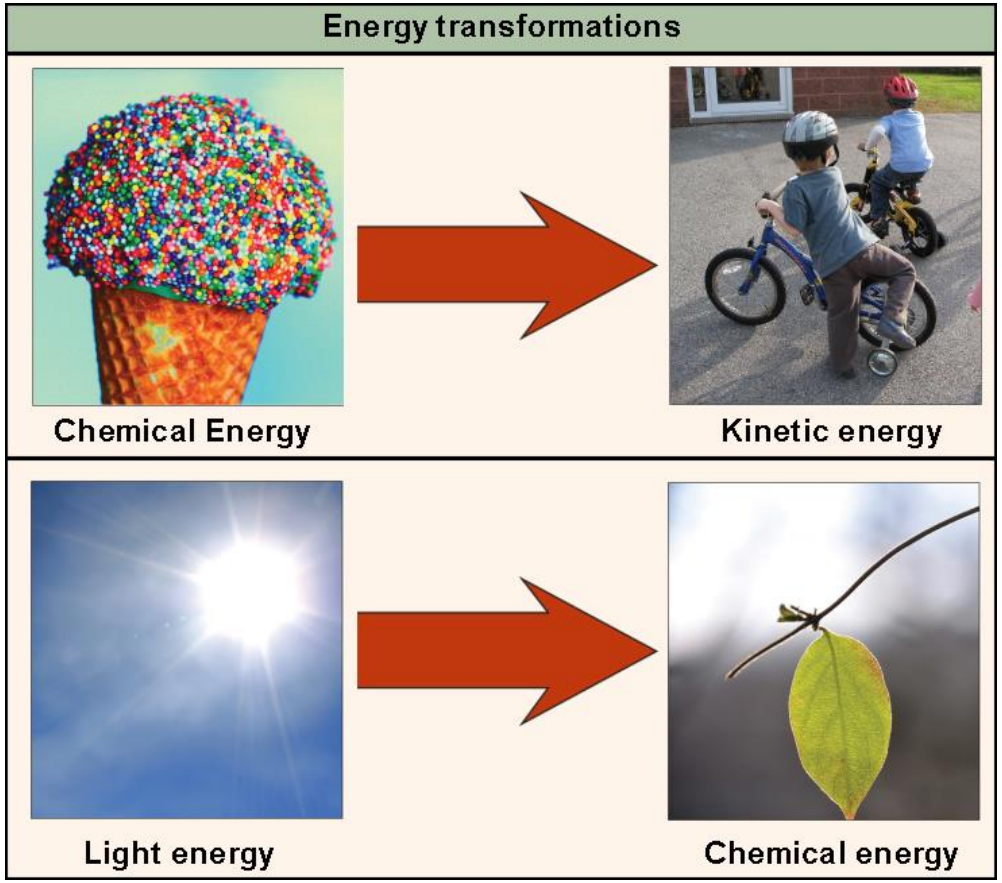
- If Body A has same temperature as body C and body B has same temperature as C, then A and B have the same temperature. This forms the basis for comparison of temperatures.
- **The Zeroth Law states that, *if two systems are in thermal equilibrium with a third system, then the two systems are equally in thermal equilibrium with each other*.**
- The third body or system is usually a thermometer.





First Law of Thermodynamics

- It obeys the **conservation of energy** which states that energy can neither be created nor destroyed but can be transformed from one form to other.
- If a system does not interact with its environment in any way then no energy comes in and no energy goes out. Energy is "conserved".
- There is always an energy balance.



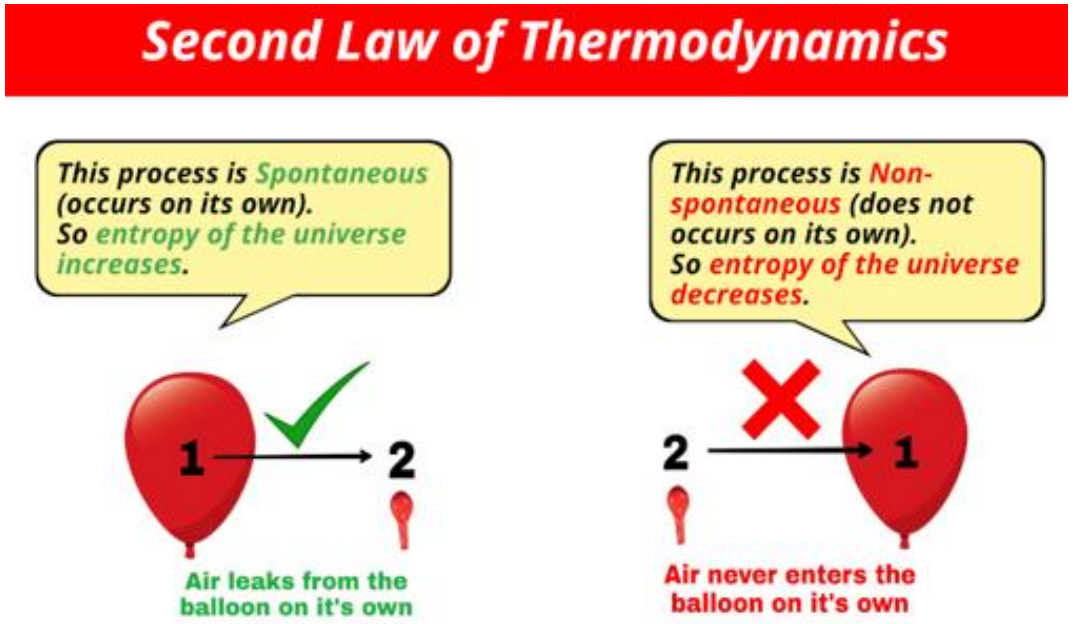
Second Law of Thermodynamics

❖ **Aspects of Second Law include;**

✓ Predicting the direction of processes.

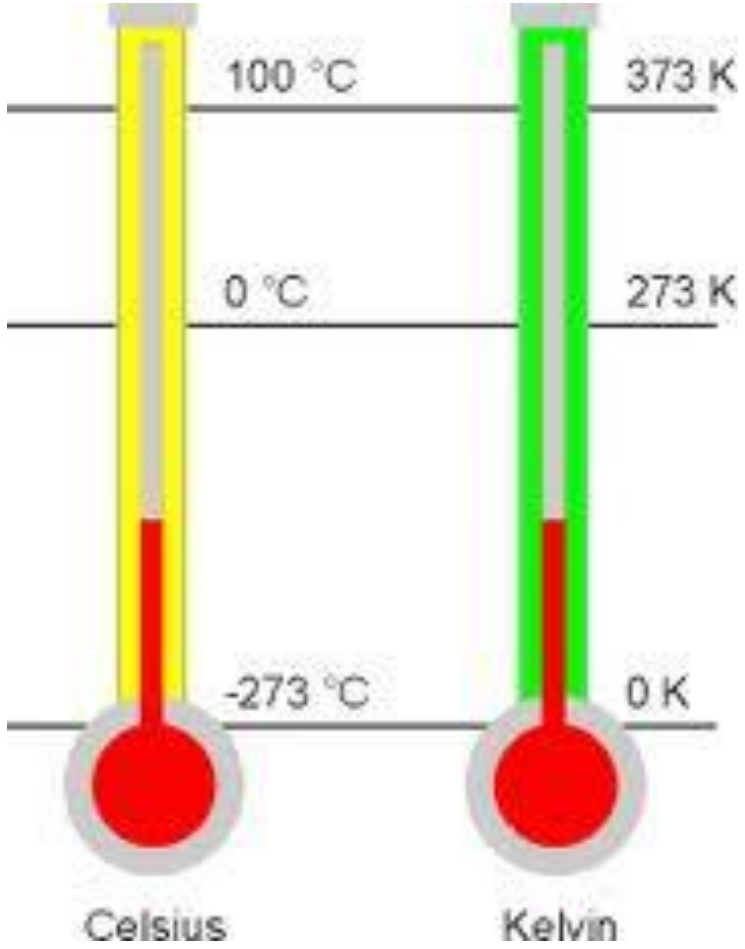
✓ Establishing conditions for equilibrium. determining the best theoretical performance of cycles, engines, and other devices.

✓ Evaluating quantitatively the factors that preclude attainment of the best theoretical performance level.



Third Law of Thermodynamics

- **It states that, it is impossible to attain a temperature of absolute zero.**
- On the Kelvin temperature scale, T = 0K is often referred to as absolute zero.
- If absolute zero temperature (0K) can be obtained, 100% efficiency can be achieved.



Applications of Thermodynamics









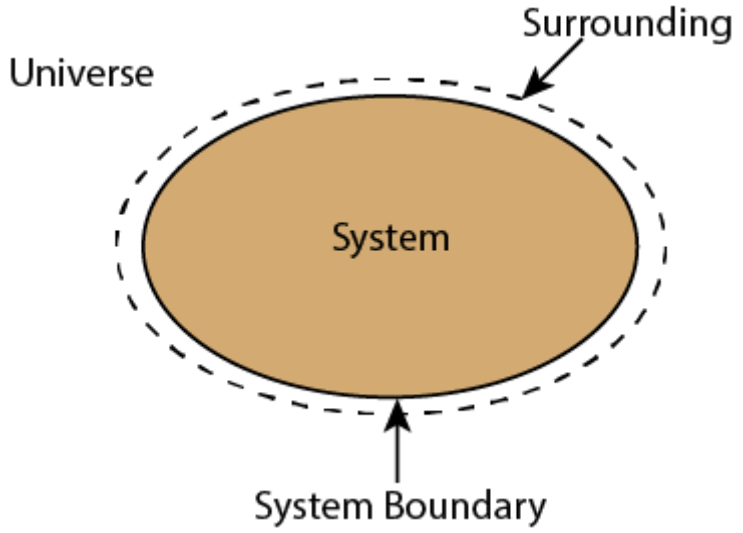






Basic Concepts and Definitions

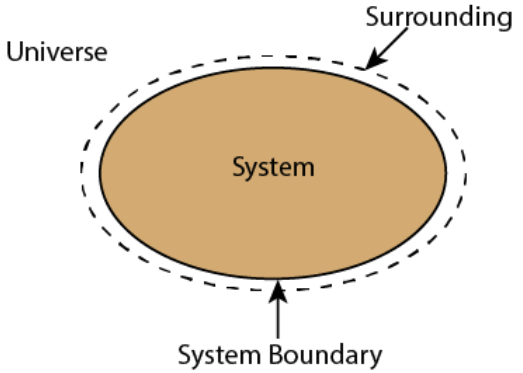
- **Thermodynamic system** is a region of space or an amount of matter, bounded by an arbitrary surface.
- **Everything external to the system is considered to be part of the system's surroundings.**
- The system is distinguished from its surroundings by a specified boundary, which may be at rest or in motion.
- A **boundary** is a closed surface, either real or imaginary, surrounding a system through which energy and mass may enter or leave the system and which separates the system from its surroundings.
- ✓ Permeable and process boundaries allow mass transfer to occur.
- ✓ A diathermal boundary allows heat transfer to occur across it as in the case of thin metal walls.



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Basic Concepts and Definitions

- Once the system is defined and the relevant interactions with other systems are identified, one or more physical laws or relations are applied.
- The system is whatever we want to study. We may want to study a quantity of matter contained within a closed, rigid-tank, or we may want to consider something such as a gas pipeline through which matter flows.



Note:

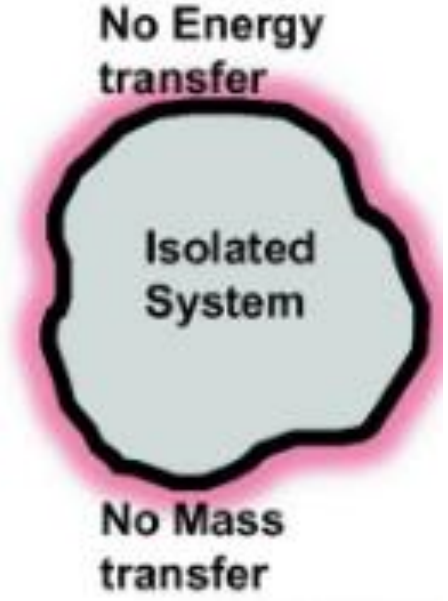
- Even vacuum which is devoid of matter, may be the focus of interest. The composition of matter inside a system may be fixed or may be changing through chemical or nuclear reactions. **The shape or volume of the system being analysed is not necessarily constant**, as when a piston compresses a gas in a cylinder or a balloon is being inflated.

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Basic Concepts and Definitions

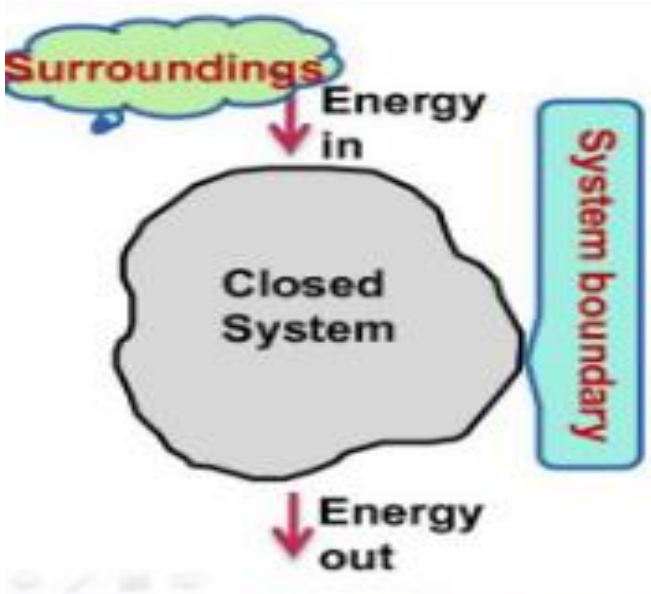
Classification of Systems

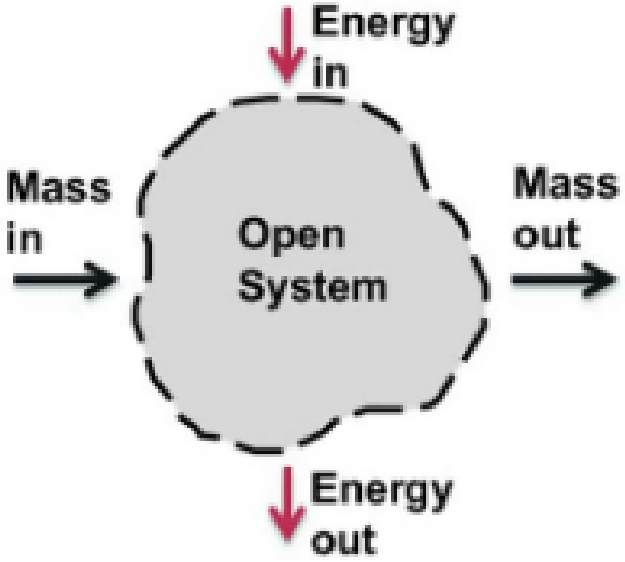
❖ A thermodynamic system interacts with its surroundings through its boundary in two ways: by exchanging energy and matter (mass).



Types of systems

- ✓ Closed system.
- ✓ Open system
- ✓ Isolated system

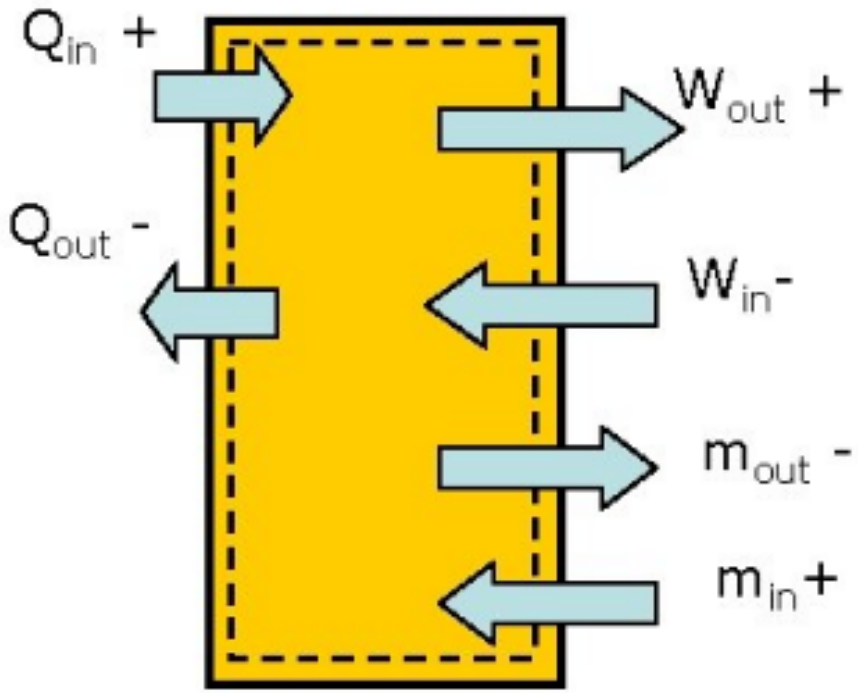




Basic Concepts and Definitions

Sign Convention

❖Energy interaction occurs in two ways; heat and work.



Sign convention

Q_{in} is positive

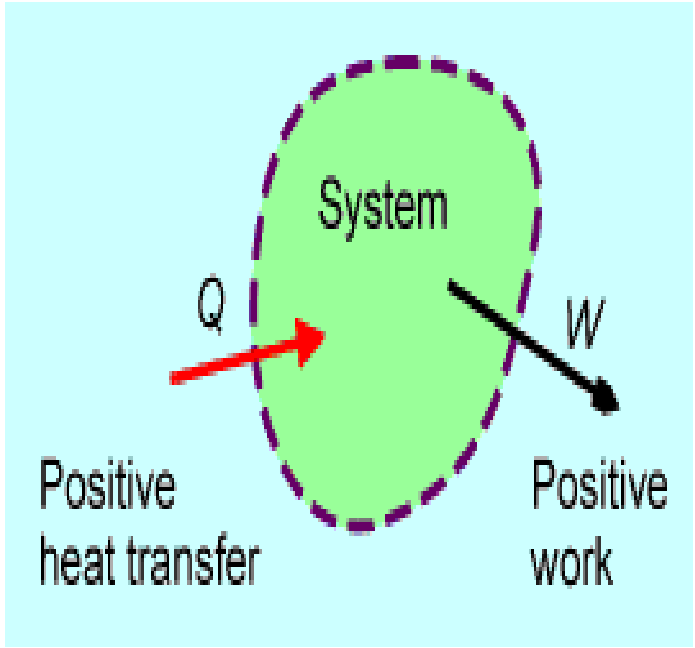
Q_{out} is negative

W_{in} is negative

W_{out} is positive

m_{in} is positive

m_{out} is negative



Basic Concepts and Definitions

Thermodynamic Properties

- A property is a characteristic of a system. It is used to specify the condition of that system.
- The most common properties are pressure, temperature, mass, volume and energy.
- A property must be macroscopic, measurable and have a single numerical value that does not depend on how the condition of the system was attained.
- **Thermodynamic properties are classified as Intensive and Extensive properties.**
- Extensive properties depend on the extent or size of the system and hence vary directly with the mass. Examples are mass, volume, energy.
- Intensive properties are independent of the size of the system. Examples are Pressure, and temperature.
- **Specific properties (denoted by lower case letters) are defined as the value of an extensive property per unit mass. They are therefore, independent of mass and are regarded as intensive properties.**

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Basic Concepts and Definitions

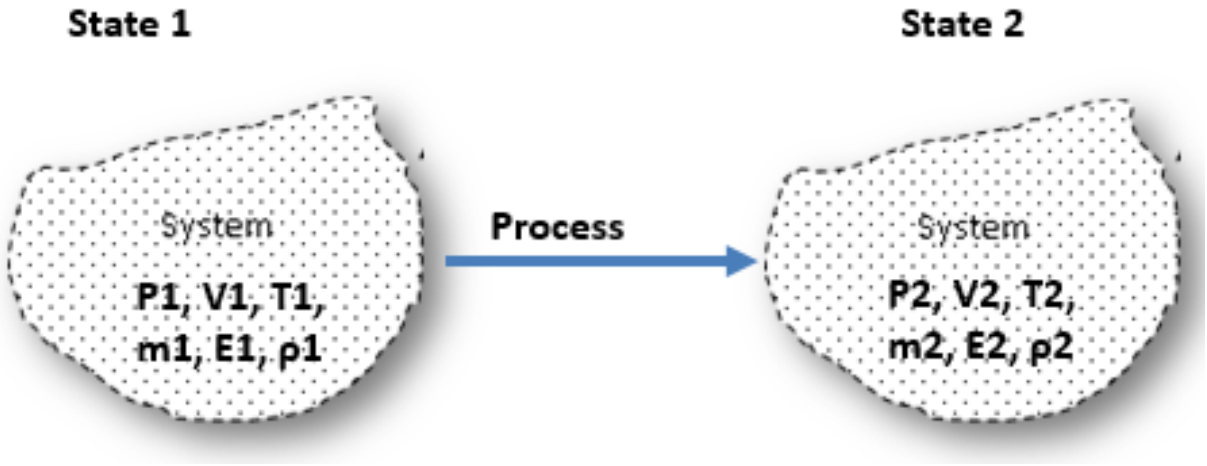
Thermodynamic State

- The word **state** refers to the condition of a system as described by its properties.
- The state can be specified by at least two thermodynamic properties.

Thermodynamic Process

- When any of the properties of a system change the state changes and the system is said to have undergone a process.
- A process is a transformation from one thermodynamic state to another.

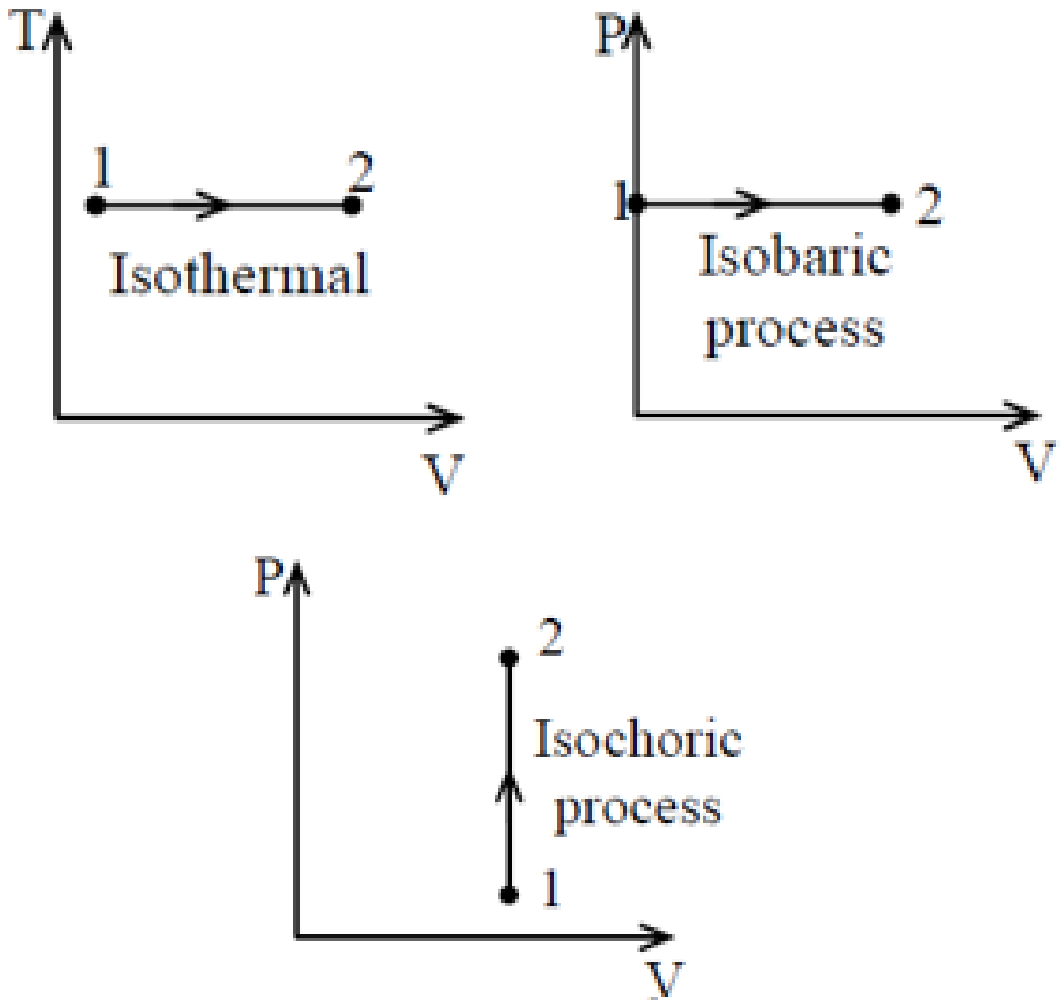
- If a system exhibits the same values of its properties at two different times, it is in the same state at those times.
- A system is said to be at steady state if none of its properties change with time.



Basic Concepts and Definitions

Path of a process

- The series of states through which a system passes during a process is called the path of the process.
- To describe a process completely, one should specify the initial and final states of the process, as well as the path it follows, and the interaction with the surroundings.
- An isothermal process occurs at constant temperature.
- An isobaric process is at constant pressure.
- An isochoric process is at constant volume.



Basic Concepts and Definitions

Point Functions

- The value of a property of a system is independent of the path of the process undergone by the system.
- It is therefore referred to as a point or state function.
- The change in the value of a state function depends solely on its values at the initial and final states and not on the path.

$$\int_1^2 dy = y_2 - y_1 = \Delta y$$

Path Functions

- Quantities such as heat and work are not properties of the system and therefore depends on the path of the process undergone by the system.
- Hence, they are referred to as path functions.
- The change in value of a path function depends on the path of the process.
- They are symbolized using inexact differential such as δQ (for heat) and δW (for work)

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Basic Concepts and Definitions

Thermodynamic cycle

- A thermodynamic cycle is a sequence of processes that begins and ends at the same state.
- In a cycle, the net change in value of any property is equal to zero.
- Cycles that are repeated periodically play prominent roles in many areas of application. For example, steam circulating through an electrical power plant executes a cycle.

Thermodynamic Equilibrium

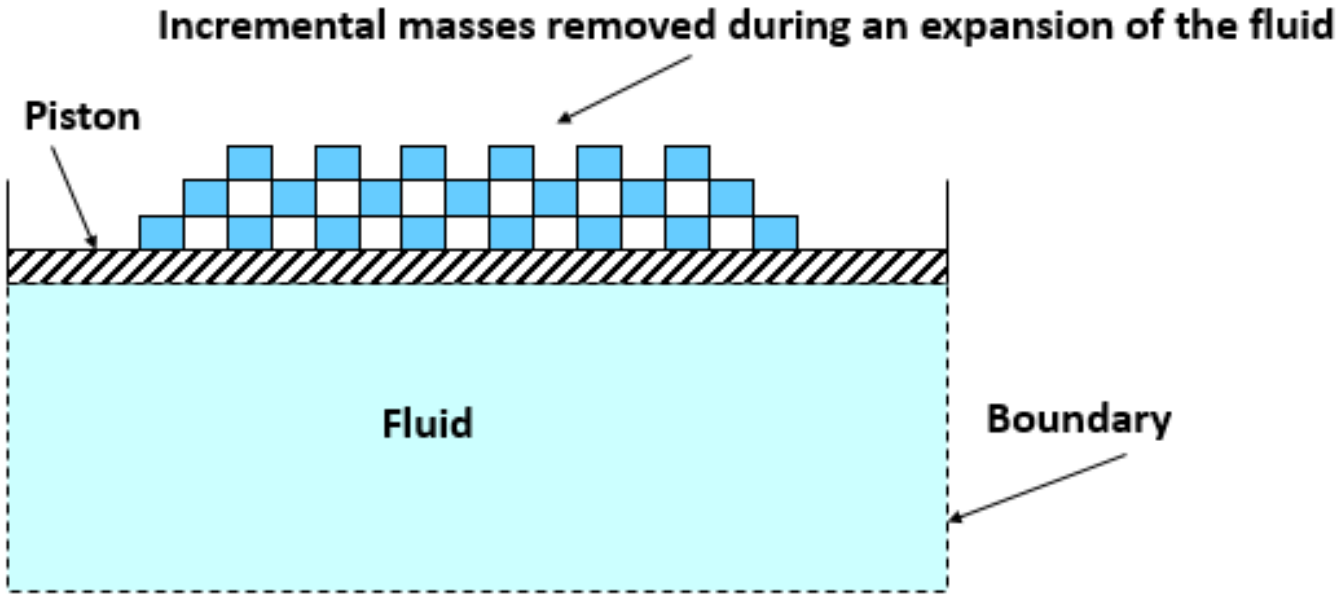
- A system is in thermodynamic equilibrium with its surroundings if no further changes occur in its properties:
- 3 conditions are satisfied:
- ✓ mechanical equilibrium: forces are balanced.
- ✓ thermal equilibrium: temperature is uniform.
- ✓ chemical equilibrium: no change in composition and no transfer of matter.

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Basic Concepts and Definitions

Quasi-Static Process

- A process is said to be quasi-static if it proceeds in such a manner that, at any instance, the system is only infinitesimally displaced from the equilibrium with the surroundings.
- A quasi-static process implies the system is essentially in internal equilibrium so that the properties are uniform.



Adiabatic Process

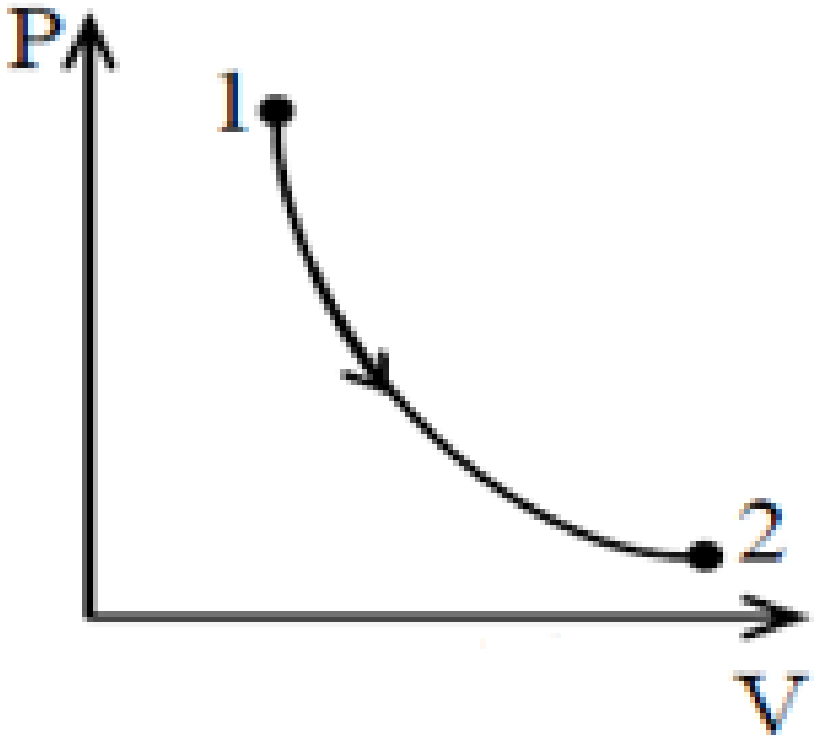
- A process during which there is no heat transfer is called an adiabatic process.
- There are two ways a process can be adiabatic.
- Thermal insulation or system having same temperature with surroundings.

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Basic Concepts and Definitions

Reversible Process

- A process is said to be reversible if after it has been carried out it is possible by any means whatsoever to restore the system and surroundings to exactly the same initial states as before.
- Since thermodynamics deals only with systems in equilibrium, it follows that it deals necessarily with reversible processes.
- **In practice, the reversible process represents the optimum path for all energy conversion processes.**
- A good example of reversible processes is the evaporation of liquid in which temperature and pressure remains essentially the same throughout the processes.
- The process path is indicated by a solid line.

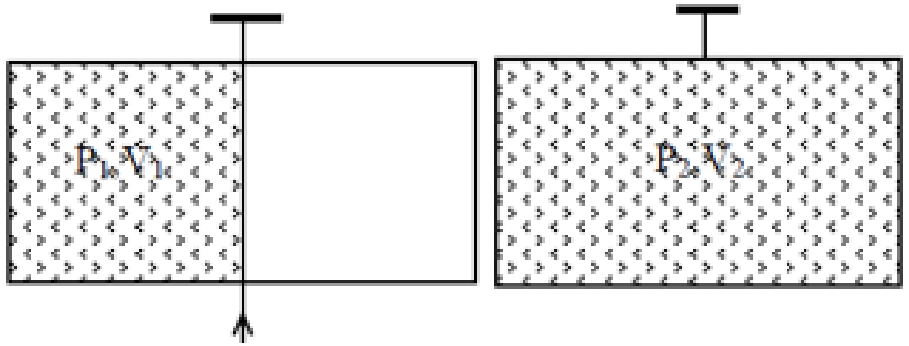


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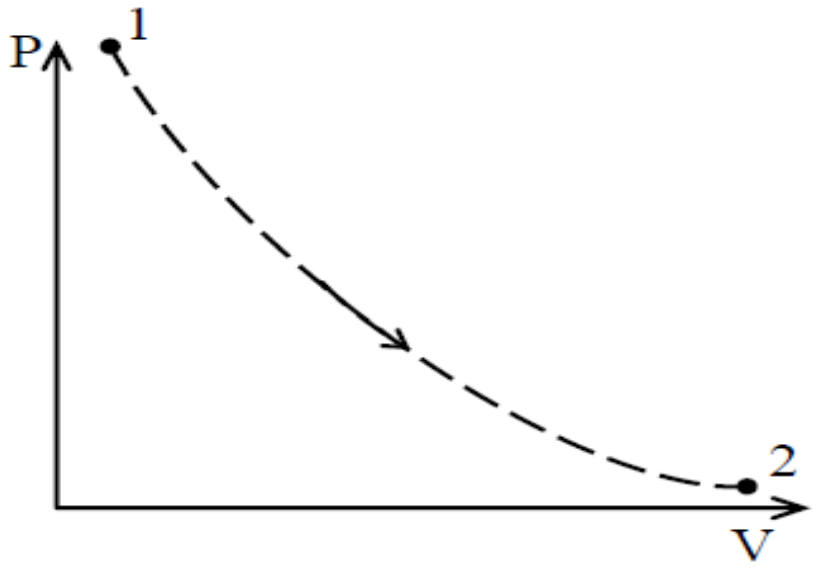
Basic Concepts and Definitions

Irreversible Process

- If the system departs significantly from equilibrium during the process, both internally and externally, the process is irreversible.
- The direction of such a process cannot be interchanged because of the differences in the values of properties within the system and also between the system and surroundings.
- Irreversible processes include doing work against friction, unrestrained expansion, mixing of two fluids, heat transfer across a finite temperature difference, inelastic deformation of solids, chemical reactions and the transfer of current against electrical resistance.
- Process path is represented by broken lines.



- **Unrestrained expansion of a gas**



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Basic Concepts and Definitions

Energy, Heat and Work

- **Energy** is simply defined as the capacity to do work or supply heat.
- **Heat** and **work** are transient forms of energy which appear only at the boundary of systems.
- **Heat** is **energy flow** by virtue of a temperature difference. Heat is not contained in a system. Heat only exists as energy crossing the boundary.
- A distinction is made between thermal energy and heat. Thermal energy is part of the internal energy, hence contained in the system. Heat is thermal energy crossing the boundary.
- **Work** is defined in mechanics as the product of force and distance, measured in the direction of application of the force.
- **Note: Work, like heat, is energy in transit and not a property of a system.**
- There are many forms: electrical, magnetic, chemical and mechanical work (including work due to expansion and compression).

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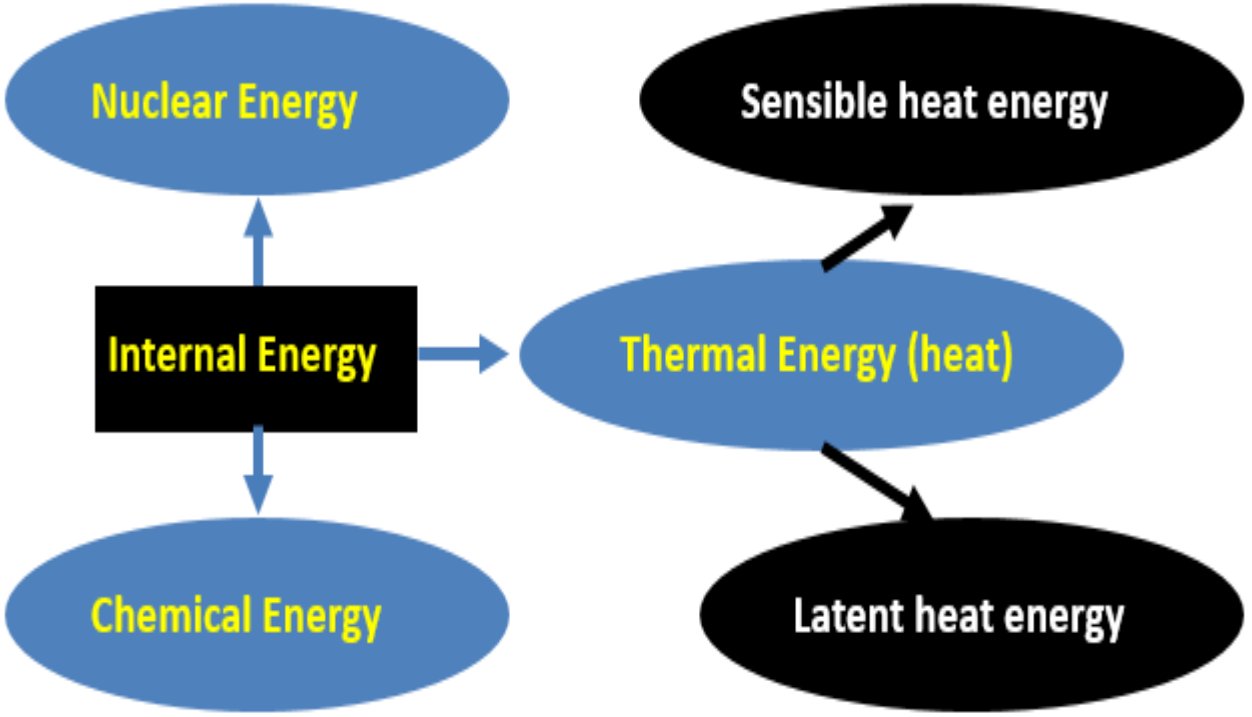
Basic Concepts and Definitions

Internal Energy, U

- **Internal Energy** is the energy associated with the microscopic components of the system.
- It includes the kinetic and potential energy associated with the random motion (rotational, vibration and translation) of the atoms and molecules.
- Also includes potential energy bonding the particles together.

Note

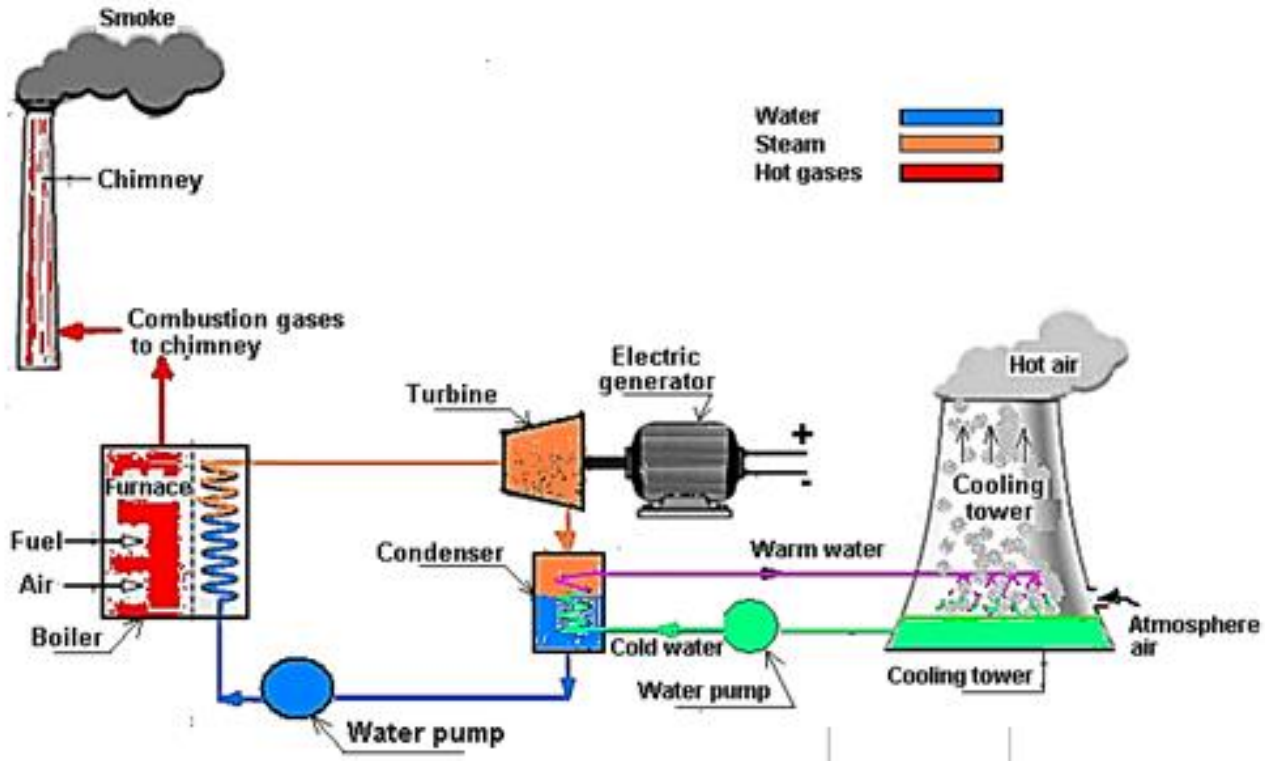
- **It does not include the macroscopic kinetic energy or external potential energy.**



Basic Concepts and Definitions

Flow Energy

- **Flow energy** is considered the energy contained in the flowing fluid.
- This is considered to be different from internal energy in thermodynamics.
- It is expressed as product of pressure and volume of the fluid.
- The **total energy** associated with any thermodynamic system is equal to its internal energy plus macroscopic kinetic energy, external potential energy and the flow energy of the fluid.
- $E = U + KE + PE + (PV)_{FE}$



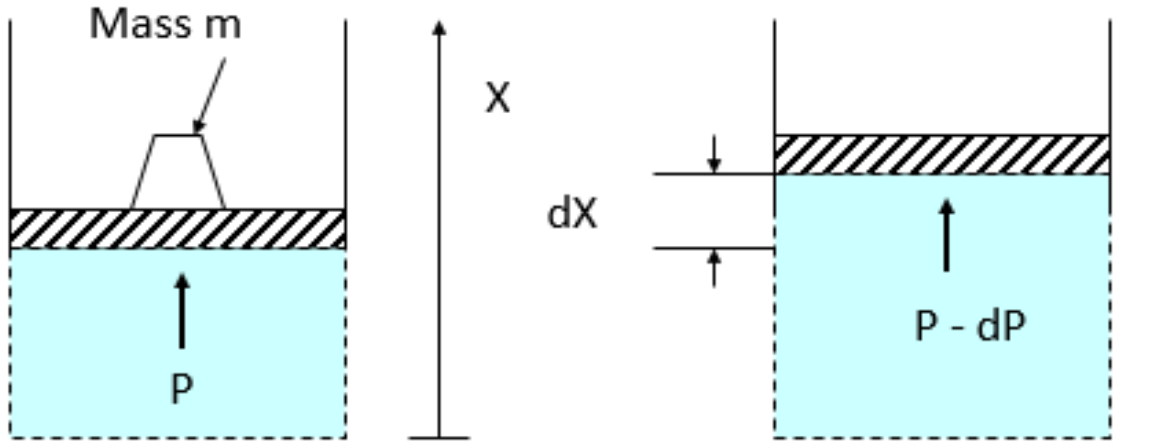
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Basic Concepts and Definitions

Simple Compressible System

- The term **simple system** is applied when there is only one form of work (either mechanical, or electrical, or magnetic, or chemical) as the system undergoes a quasi-static process.
- When **expansion** or **compression** is the only mode of work transfer, the system is said to be a simple compressible system.
- The boundary work due to expansion or compression is expressed as;

$$\delta W = PdV$$



a) Before expansion **b)After expansion**

Expansion work (Boundary work)

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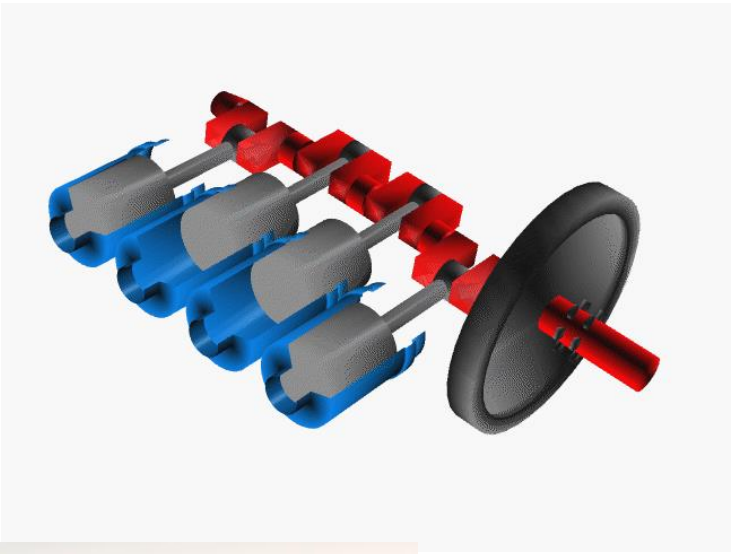
Basic Concepts and Definitions

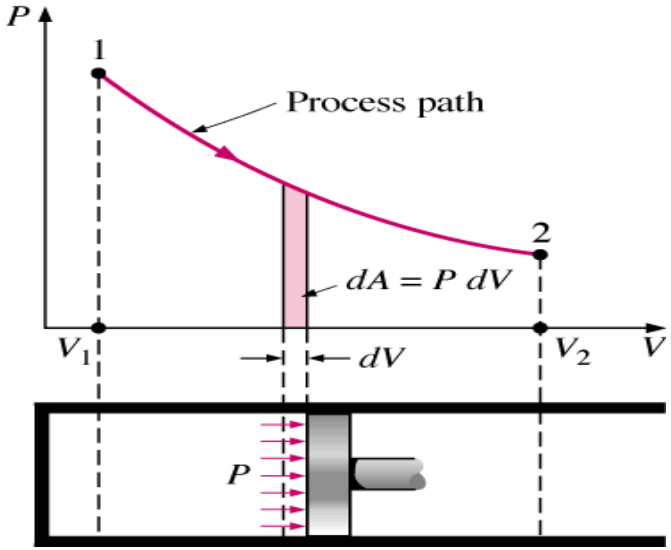
Simple Compressible System

- The boundary work due to expansion or compression is expressed as;

$$\delta W = PdV$$

- PdV work is commonly encountered in reciprocating devices such as automotive engines and compressors.










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